d) Everyone in your class enjoys Thai food.

$T(x)$: Person x enjoys Thai food

Domain: Students of your class $\forall x\, T(x)$

Domain: Students of your school $\forall x(C(x) \to T(x))$

Domain: All people $\forall x(C(x) \to T(x))$

$F(x,y)$: Person x enjoys Food y.

Domain: Students of your class $\forall x\, F(x, \text{Thai})$

Domain: Students of your school $\forall x(C(x) \to F(x, \text{Thai}))$

Domain: All people $\forall x(C(x) \to F(x, \text{Thai}))$

e) Someone in your class does not play hockey.

$H(x)$: Person x plays hockey

Domain: Students of your class $\exists x \neg H(x)$

Domain: Students of your school $\exists x(C(x) \land \neg H(x))$

Domain: All people $\exists x(C(x) \land \neg H(x))$

$G(x,y)$: Person x plays game y

Domain: Students of your class $\exists x \neg G(x, \text{Hockey})$

Domain: Students of your school $\exists x(C(x) \land \neg G(x, \text{Hockey}))$

Domain: All people $\exists x(C(x) \land \neg G(x, \text{Hockey}))$

29) Express each of these statements using logical operators, predicates and quantifiers.

a) Some propositions are tautologies.

Domain for question 29: Set of all statements

$P(x)$: x is a proposition $T(x)$: x is a tautology

$C(x)$: x is a contradiction $N(x,y)$: x is the negation of y

$Co(x)$: x is a contingency

a) $\exists x(P(x) \land T(x))$

b) The negation of a contradiction is a tautology

$\forall x(C(x) \land N(y,x) \Rightarrow T(y))$